

## **Writeup. J.P. Morgan Macrosynergy Quantamental System Bond Yield Forecasting Economic Rationales**

Bond returns carry a small but real premium that varies with economic cycles, which means part of the return is forecastable from macro fundamentals. The opportunity is to capture that premium systematically across many markets at once. In a J.P. Morgan Macrosynergy competition, I set out to forecast weekly bond excess returns across nine developed markets from macro indicators. The hard part of this forecasting problem is that the target is constructed at weekly frequency while its economic drivers usually change at a slower time horizon. To see why this matters, we decompose the weekly bond excess return into two parts. One is its conditional mean given all information known today, which is economically the risk premium an investor is compensated to bear. The other is the part no information available today could have predicted, such as the week's shock from news and trading flows.

The thesis is that the predictable component of weekly bond excess returns comes from slow-moving level signals with an economic reason behind them, chiefly carry and expected inflation, which proxy the risk premia, and that this mechanism-grounded view holds consistently across developed markets, with a higher information coefficient than signals built from changes and surprises. The claim indicates two types of robustness worth separating: breadth across markets, and stability across time. Carry is essentially a slope measure of the yield curve. It equals roughly the difference between expected future short interest rates and today's short rate plus a term premium that compensates for duration risk. If a steep yield curve only reflects expected interest rate hikes, it would predict no excess returns, because the coming rate rise would push the bond price down by the same amount as the carry collected. Empirically, a steep curve predicts positive excess returns, which means much of the slope reflects the term premium rather than expected hikes. The inflation channel works through the Fisher identity, which decomposes the nominal yield into a real yield, an expected inflation term, and an inflation risk premium. So carry and expected inflation each proxy a different piece of the compensation an investor earns for holding the bond, the term premium and the inflation risk premium respectively.

The model reached about 58 percent out-of-sample directional accuracy overall, a modest edge whose source is worth examining, which I return to below. The first half of the thesis held: the level signals were robust across markets, keeping the same sign in seven or eight of the nine markets depending on the signal, with no single market driving the result. The second half did not. When the data is split chronologically, the relationships that held through the low-inflation period before 2022 broke in the 2022-2023 inflation shock, with several coefficients flipping to signs that contradict economic theory. The research I contributed to on bond-stock comovement explains the reversal: inflation's effect on bonds depends on the type of economic shocks that drive it and how monetary policy responds. Demand-driven inflation is procyclical, meaning it rises in expansions and falls in recessions, so bonds pay off when growth weakens and behave like a hedge. Conversely, supply-driven inflation is countercyclical, which, together with a strong anti-inflation policy response, can produce high-inflation recessions, making bonds risky. So

different inflation dynamics before and after 2022 pull the same coefficient in opposite directions, and a pooled model averages them into a sign that fits neither.

A second lesson came from a mistake of my own. Trying to make the model see where it was in the rate cycle, I added an interest-rate-cycle feature, and performance jumped sharply, until a forward-lag test showed that the gain was from data leakage. The feature was dated to the end of the week being forecast. Once it was lagged properly, out-of-sample Root Mean Square Error (RMSE) skill fell from about ten percent to around three percent, meaning the model's forecast error was only about three percent lower than a zero-return forecast. So I discarded it and kept the more conservative result. More broadly, an edge one can explain is worth more than an edge that merely fits, because a mechanism survives regime shifts while a correlation does not.

What I would do differently splits into research discipline and what to do with an unstable signal in practice. On research discipline, the forward-lag leakage test belongs at the feature-construction stage before fitting the model, so a feature has to survive a proper lagging check before it can be used to construct any signal. Additionally, a time-series split is not sufficient, because it prevents ordering leakage but not within-week feature-timing leakage. The missing guard is a point-in-time audit: every feature must use only information available as of the prediction date. But improving the research is only half of the job. In practice, the more important question is what to do once a signal is known to be unstable. What I take from this is to treat instability as information: lower conviction, smaller size, and more weight on the signals that survive across regimes. The largest remaining gap is between predictability and investability. The 58 percent reflects the common direction of developed-market bonds, which tend to rise and fall together, rather than skill at selecting which market outperforms the others. A statistical result is not yet a tradable one. It still needs to be measured net of transaction costs, turnover, and capacity to evaluate whether the signal is tradable, since a signal that trades this often can have its edge consumed entirely by costs.